LOCAL NEVANLINNA MEMBERSHIP BELOW TWO

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Are elements of $\mathscr{H}^p$ for $p < 2$ locally in the Nevanlinna class?

Saksman and Seip, *Some open questions in analysis for Dirichlet series*, arXiv:1601.01616, Problem 2.4.

Answered: no. Evidence: Sketch.

No. The same function as above has local logarithmic means diverging along a sequence of vertical lines.

Proof. Let $A = 64$, $a_k = A^k$, $N_k = \lceil \exp(\exp(A^{8k})) \rceil$, $L_k = \log N_k$, $R = A^4$, and

$$F(s) = \sum_{k \geq 1} \frac{a_k}{\sqrt{N_k}} P_{N_k}(s), \qquad P_N(s) = \sum_{n \leq N} n^{-s}.$$

Harper's *Moments of random multiplicative functions, I: Low moments, better than squareroot cancellation, and critical multiplicative chaos*, Theorem 1, doi:10.1017/fmp.2019.7, proves $F \in \mathscr{H}^p$ for every $p < 2$, and the coefficient series converges absolutely in the half-plane. For fixed T, sum-integral comparison uniformly for $|t| \leq T$ gives

$$P_N(\sigma + it) = \frac{N^{1-\sigma-it}}{1-\sigma-it} + O_T(1).$$

At $s_k = 1/2 + L_k^{-1} + it$, the kth main numerator in $(1 - s_k)F(s_k)$ has modulus a_k/e. Earlier terms sum to at most $a_k/63$; later terms sum to less than $a_k/100$ since $L_{k+m}/L_k \geq R^m$ and $R = A^4$. The accumulated sum-integral error is bounded independently of k. Since $|1 - s_k| \leq (1/4 + T^2)^{1/2}$, it follows for all large k that

$$\inf_{|t| \leq T} |F(1/2 + L_k^{-1} + it)| \geq c_T A^k.$$

Thus the integral of $\log^+ |F|$ over every fixed interval of positive length tends to infinity along L_k^{-1}. If F were locally in the classical Nevanlinna class, a positive harmonic majorant on a rectangle adjoining a smaller boundary interval would dominate these integrals: the bottom harmonic-measure density at a fixed interior point has a uniform positive lower bound on that interval. This contradicts divergence. □

AI assistance was used in adapting and checking the argument and the exposition.

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